

Gelei Xu

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Research Interests

Keywords: Trustworthy AI for Healthcare, Medical Image Analysis, AI Agents, Edge AI

My research designs population-aware machine learning methods that optimize clinical AI for targeted patient groups across data selection, model adaptation, and system evaluation. I am currently extending this work to continually learning systems and evidence-bound healthcare agents.

Education

University of Notre Dame

Ph.D. Student in Computer Science and Engineering
Advisor: Yiyu Shi

2023–Present
Notre Dame, IN, USA

Southern University of Science and Technology

B.E. in Computer Science and Engineering
Advisor: Jiang Liu

2019–2023
Shenzhen, Guangdong, China

Professional Experience

Weill Cornell Medicine

Visiting Graduate Student
Host: Fei Wang

Aug 2026–Present
New York, NY, USA

Boston Children's Hospital & Harvard Medical School

Machine Learning Intern
Host: Maimuna Majumder

Jun 2026–Aug 2026
Boston, MA, USA

Journal Publications

J.5 Assessing the Use of Automatic Speech Recognition and Large Language Models for Individuals with Language Impairments

Gelei Xu^{*}, Haoxinran Yu^{*}, Lixuan Wei, Yuxuan Liu, Dancheng Liu, Chenhui Xu, Jiajie Li, Ahmed Abbasi, Jinjun Xiong, Xiufan Yu, Zhi Zheng, Yiyu Shi, and Ruiyang Qin
Nature Communications, 2026 (*IF*=18.1)

J.4 Rethinking Fairness in Medical Imaging: Maximizing Group-Specific Performance with Application to Skin Disease Diagnosis

Gelei Xu, Yuying Duan, Jun Xia, Ching-Hao Chiu, Michael Lemmon, Wei Jin, and Yiyu Shi
Medical Image Analysis, 2026 (*IF*=14)

J.3 A Survey of AI Agents in Healthcare

Gelei Xu*, Xueyang Li*, Yixiong Chen*, Yuying Duan*, Shuqing Wu*, Alexander Yu*, Ching-Hao Chiu*, Juntong Ni*, Ningzhi Tang, Toby Jia-Jun Li, Alan Yuille, Wei Jin, and Yiyu Shi

Journal of Biomedical Informatics, 2026 (*JBIM's most downloaded articles, [GitHub 1.2k stars](#)*)

J.2 Automatic Cortical Cataract Classification Framework Based on AS-OCT Images

Gelei Xu*, Xiaoqing Zhang*, Zunjie Xiao, Risa Higashita, Wan Chen, Jin Yuan, and Jiang Liu

Computer Systems & Applications, 2022

J.1 CCA-Net: Clinical-awareness Attention Network for Nuclear Cataract Classification in AS-OCT

Xiaoqing Zhang, Zunjie Xiao, Lingxi Hu, Gelei Xu, Risa Higashita, Wan Chen, Jin Yuan, Jiang Liu

Knowledge-Based Systems, 2022

Conference Publications**C.12 Group-Conditioned Representation Modulation for Fair Skin Disease Diagnosis**

Gelei Xu*, Haoran Fan*, Yiyu Shi

Proceedings of the 2026 Medical Image Computing and Computer Assisted Interventions (MICCAI 2026, early acceptance, acceptance rate 9%)

C.11 Beyond Visual Forensics: Auditing Multimodal Robustness for Synthetic Medical Image Detection

Ching-Hao Chiu, Hao-Wei Chung, Gelei Xu, Xueyang Li, Pin-Yu Chen, John Kheir, Meysam Ghaffari, Carlos Morato, Ahmed Abbasi, Yiyu Shi

Proceedings of the 2026 Medical Image Computing and Computer Assisted Interventions (MICCAI 2026)

C.10 MORI-Seg: Learning Morphological Geometry for Instance Segmentation without Instance Annotations

Leiyue Zhao, Tianyu Shi, Daniel Reisenbuechler, Xinzi He, Junchao Zhu, Tianyuan Yao, Yuechen Yang, Yanfan Zhu, Junlin Guo, Gelei Xu, Haichun Yang, Yuankai Huo, Mert R. Sabuncu, Yihe Yang, Ruining Deng

Proceedings of the 2026 Medical Image Computing and Computer Assisted Interventions (MICCAI 2026, early acceptance, acceptance rate 9%)

C.9 How Coding Agents Fail Their Users: A Large-Scale Analysis of Developer-Agent Misalignment in 20,574 Real-World Sessions

Ningzhi Tang, Chaoran Chen, Gelei Xu, Yiyu Shi, Yu Huang, Collin McMillan, Tao Dong, Toby Jia-Jun Li

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)

C.8 NaturalEdit: Code Modification through Direct Interaction with Adaptive Natural Language Representation

Ningzhi Tang, David Meininger, Gelei Xu, Yiyu Shi, Yu Huang, Collin McMillan, Toby Jia-Jun Li

Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST 2026)

C.7 Programming by Chat: A Large-Scale Behavioral Analysis of 11,579 Real-World AI-Assisted IDE Sessions

Ningzhi Tang*, Chaoran Chen*, Zihan Fang, Gelei Xu, Maria Dhakal, Yiyu Shi, Collin McMillan, Yu Huang, Toby Jia-Jun Li

Proceedings of the 2026 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)

C.6 Tiny-Align: Bridging Automatic Speech Recognition and Large Language Model on Edge

Ruiyang Qin, Dancheng Liu, Gelei Xu, Amir Nassereldine, Zheyu Yan, Chenhui Xu, Yuting Hu, Shaocong Wang, Xiaobo Sharon Hu, Jinjun Xiong, Yiyu Shi

Proceedings of the 2025 International Conference on Computer-Aided Design (ICCAD 2025, acceptance rate 24.7%)

C.5 Incorporating Rather Than Eliminating: Achieving Fairness for Skin Disease Diagnosis Through Group-Specific Experts

Gelei Xu, Yuying Duan, Zheyuan Liu, Xueyang Li, Meng Jiang, Michael Lemmon, Wei Jin, Yiyu Shi
Proceedings of the 2025 Medical Image Computing and Computer Assisted Interventions (MICCAI 2025, early acceptance, acceptance rate 9%)

C.4 The Cost of Local and Global Fairness in Federated Learning

Yuying Duan, Gelei Xu, Yiyu Shi, Michael Lemmon
Proceedings of the 2025 International Conference on Artificial Intelligence and Statistics (AISTATS 2025)

C.3 Enabling On-Device Learning via Experience Replay with Efficient Dataset Condensation

Gelei Xu, Ningzhi Tang, Jun Xia, Ruiyang Qin, Wei Jin, Yiyu Shi
Proceedings of the 2024 Design, Automation & Test in Europe Conference & Exhibition (DATE 2024, acceptance rate 25%)

C.2 A Novel Local-Global Spatial Attention Network for Cortical Cataract Classification in AS-OCT

Zunjie Xiao, Xiaoqing Zhang, Qingyang Sun, Zhuofei Wei, Gelei Xu, Yuan Jin, Risa Higashita, Jiang Liu
Proceedings of the Pattern Recognition and Computer Vision: 5th Chinese Conference (PRCV 2022)

C.1 Channel-Wise and Spatial Feature Recalibration Network for Nuclear Cataract Classification

Xiaoqing Zhang, Gelei Xu, Junyong Shen, Zunjie Xiao, Qiuyang Yan, Jin Yuan, Risa Higashita, Jiang Liu
Proceedings of the 2022 IEEE International Conference on Multimedia and Expo (ICME 2022)

Workshop Papers

W.2 Achieving Fairness in Dermatological Disease Diagnosis Through Automatic Weight Adjusting Federated Learning and Personalization

Gelei Xu, Yawen Wu, Zhengge Jia, Jingtong Hu, Yiyu Shi
Proceedings of the MICCAI 2025 Workshop on Fairness of AI in Medical Imaging (FAIMI)

W.1 An Adaptive System for Wearable Devices to Detect Stress Using Physiological Signals

Gelei Xu, Ruiyang Qin, Zhi Zheng, Yiyu Shi
2024 Workshop on Conference on Human Factors in Computing Systems (CHI)

Selected Honors and Awards

IDEA Center Edison Innovation Fellowship (\$16,500)	2025–2026
MICCAI 2025 NIH Registration Grant (\$700)	2025
Downes Memorial Fund (\$1000)	2025
Zahm Professional Development Fund (\$5250)	2024–2026
DAC Young Fellow	2024
Lyman Taylor & W Stuart Travel Award (\$1500)	2024
Excellent Graduate of SUSTech, CSE Department, and Zhicheng College	2023
Enterprise Intelligence Base Scholarship (¥5000)	2022

Teaching Experience

Graduate Associate for Teaching 2025–2026
 Kaneb Center for Teaching Excellence, University of Notre Dame
Instructor, *Foundations of Teaching Workshop Series*
Instructor, *Assessing Prior Knowledge Workshop*
Blog Post Author ([link](#)), *How AI Can Make More Room for Human Connection in Teaching*
Guest Lecturer, *EE 60572 Introduction to Deep Learning* Spring 2025
 Department of Electrical Engineering, University of Notre Dame *Host: Michael Lemmon*
Teaching Assistant, *CSE 20221 Logic Design* Spring 2024
 Department of Computer Science and Engineering, University of Notre Dame
Teaching Assistant, *CSE 40535 Computer Vision* Fall 2023
 Department of Computer Science and Engineering, University of Notre Dame

Professional Service

Workshop Organizer

EMERGE @ MICCAI 2026: Empowering Medical Information Computing and Research through
 Early-career Guidance and Expertise *Strasbourg, France*
Gelei Xu, Naren Akash, Cosmin I. Bercea

FAIMI @ MICCAI 2026: Fairness of AI in Medical Imaging *Strasbourg, France*
 Aasa Feragen, Andrew King, Ben Glocker, Enzo Ferrante, Eike Petersen, Esther Puyol-Antón, Melanie
 Ganz-Benjaminsen, Veronika Cheplygina, Louisa Fay, Emma Stanley, Gelei Xu, Christopher Thomas
 Boland, Tareen Dawood

Reviewing

Journal: JBHI 2026, CSUR 2026, BMJEBM2026, TCPS 2025, TNSRE 2025, Scientific Reports 2025,
 TOIS 2024, TECS 2024, Scientific Reports 2024
Conference: AAAI 2026, MICCAI 2026, RO-MAN 2025, MIDL 2025, MICCAI 2025, RO-MAN 2024
Workshop: MICCAI EMERGE 2026, MICCAI FAIMI-BRIDGE-EPIMI 2026, MICCAI Educational
 Challenge 2026, KDD 2026 SciSoc Agents and LLMs, AAAI Fall Symposium Series 2024

Outreach

Scientific Events Officer, MICCAI Student Board, 2026
Graduate Recruitment Ambassador, Notre Dame College of Engineering, NDiSTEM 2025
Invited Panelist, Undergraduate Outreach Program, Notre Dame College of Engineering, 2025
Mentor Judge, NDiSTEM 2025 Research Presentations